Answer **all** the questions in this section.

1 A man travels on a toboggan down a slope covered with snow from point A to point B. The path is illustrated in Fig. 1.1.

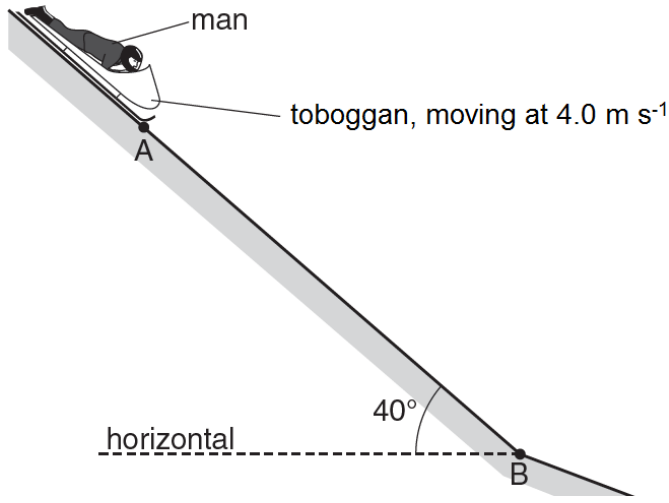

Fig. 1.1 (not to scale)

The slope AB makes an angle of 40˚ with the horizontal. Friction is **not negligible**.

The man and toboggan have a combined mass of 95 kg.

The mass starts with a speed of 4.0 m s^{-1} down the slope at A and has constant acceleration between A and B. The mass takes 17.0 s to reach B. His speed is 37.0 m s^{-1} at B.

(a) **(i)** Calculate the distance moved from A to B.

distance = ... m [2]

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(ii) The actual uncertainty in each measurement of the speed of the mass is ± 0.1 m s^{-1}. The percentage uncertainty in the time taken to move from A to B is ± 0.6%.

Calculate the percentage uncertainty in the distance moved from A to B.

percentage uncertainty = ……. % [2]

(iii) Use your answers in **(a)(i)** and **(a)(ii)** to determine the value of distance moved from A to B, with its actual uncertainty, to an appropriate number of significant figures.

distance = ……. ± ……. m [1]

(b) For the man and toboggan moving from A to B,

(i) calculate the change in kinetic energy

change in kinetic energy = ……. J [2]

(ii) calculate the change in potential energy

change in potential energy = ……. J [2]

(iii) hence determine the average frictional force that acts on the toboggan between A and B.

frictional force = ……. N [2]